Department of Chemistry
Jahangirnagar University
Savar, Dhaka 1342

Practical Manual for Chem: 315

Contents

EXPERIMENT 1
To determine the molecular weight of a substance by cryoscopic method 2

EXPERIMENT 2
Study of the adsorption of acetic acid on charcoal and examine the validity of Freundlich's adsorption isotherm and Langmuirs adsorption isotherm. 4

EXPERIMENT 3
To study the liquid-liquid miscibility of three liquid system and to determine the position of one tie line..6

EXPERIMENT 4
To determine the partial molal volume of solutions (Alcohol-Water system)by Westphal balance -------8

EXPERIMENT 5
To determine the partial molal volume of solutions (Alcohol-Water system) by pycnometer----------11

Experiment 6
To find (a) the specific rotation of sucrose (b) concentration of an unknown sucrose solution (c) Influence of solvent on the rotation of sucrose solution by polarimetric method 13

EXPERIMENT 7
To determine (a) the absorption curve of coloured substances in solution, (b) the validity of Beer-Lambert law, and (c) the strength of an unknown solution 15

EXPERIMENT 8
To determine the specific reaction rate of base catalysed hydrolysis of an ester by conductometric method 17

EXPERIMENT 9
To find out the equivalent conductance at infinite dilution and the dissociation constant of a weak electrolyte. 18

EXPERIMENT 10
Conductometric titration of (a) strong and (b) weak acids and to draw their neutralisation curves.....20

EXPERIMENT 11
Potentiometric titration of (a) strong and (b) weak acids and to draw their neutralisation curves… ------ 22

EXPERIMENT 12
To determine the standard electrode potential of zinc and copper..... 24

Scanned with CamScanner

Experiment 1

To determine the molecular weight of a substance by cryoscopic method

Apparatus and Chemicals required

Digital thermometer (−10° to 100°C), stirring rod, ice, cyclohexane, naphthalene, camphor, stop watch.

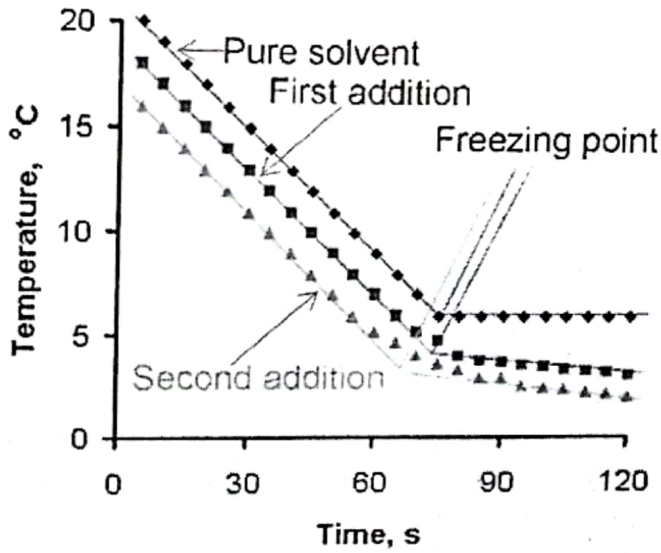
Theory: The properties of a solution differ from those of a pure solvent due to interactions that take place between the solute and solvent molecules. The properties that exhibit such changes are called the colligative properties and include vapour pressure lowering, boiling point elevation, freezing point depression and changes in osmotic pressure. These properties are dependent only upon the numberof particles (ions or molecules) that are dissolved in the solvent and not on the identity of the particles. This experiment will examine the phenomenon of freezing point depression. When a particular solute is dissolved in a solvent, the following expression holds true:

$$\Delta T = T_f^0 - T_f = K_f m = K_f \frac{1000\, w_1}{M\, w_2} (1), \quad \text{or. } M = K_f w_1 1000 / \Delta T\, w_2$$

w_1 = wt. of solute

w_2= wt. of solvent

The terms $T_f°$ and $T_f$refer to the freezing-point temperatures of the pure solvent and the solution, respectively. The term "m" indicates the molality of the solution, which is defined as the number of moles of solute per 1000 g of solvent. This quantity is used, rather than molarity, because it is not temperature dependent. The constant, K_f, is referred to as the freezing-point-depression constant and is dependent only upon the solvent. For cyclohexane, K_f= 20.2 deg·kg/mole of solute.



Scanned with CamScanner

Procedure

1. Set up the apparatus ensuring that the stirrer does not touch the thermometer.
2. Take 20-25 ml of cyclohexane into the tube and place in the air jacket and dip it in the freezing mixture. Cool it by direct immersion in the ice bath until the cyclohexane starts to freeze. Take thermometer reading at every 30 second.
3. Stir the cyclohexane slowly and read the thermometer until the constant reading is obtained.
4. Take out the freezing tube and warm it with hand until the solid melts.
5. Add measured quantity (about 0.5-0.6 gm) of the solute into the tube and dissolve it by stirring.
6. Determine the freezing point in exactly the same manner as before.
7. Plot temperature against time for solvent and solution in the same graph paper.
8. Calculate the molecular weight of the substance by using the above equation (1).

Scanned with CamScanner

Experiment 2

Study of the adsorption of acetic acid on charcoal and examine the validity of Freundlich's adsorption isotherm and Langmuirs adsorption isotherm.

Material required : Burette, pipette, activated charcoal, 0.5 N acetic acid, 0.1 N sodium hydroxide, stoppered bottles, shaking machine, phenolphthelien indicator.

Theory : The amount adsorbed by any adsorbent is dependent on pressure and temperature. Thus the amount (X) adsorbed is a function of pressure (P) and temperature (T), i.e.

$$X = f(P, T)$$

A plot of P and X, keeping temperature constant is known as adsorption isotherm.

Freundlich adsorption isotherm :Freundlich adsorption isotherm ; represented by

$$\frac{x}{m} = kc^{1/n}$$

where X is the amount of salute adsorbed, m ; the amount of adsorbing material, C is the equilibrium concentration of adsorbate in the solution, k is a constant depending upon the nature of both adsorbent and adsorbate, n is another constant dependent upon the nature of the adsorbate. The value of 1/n is generally less than unity. On taking logarithm of eqn. (1),

$$\log \frac{x}{m} = \log k + \frac{1}{n} \log c$$

If a plot of $\log \frac{x}{m}$ against log c gives a straight line with a slope of $\frac{1}{n}$ and intercept of log k. Freundlich's adsorption isotherm is valid.

Langmuir adsorption isotherm : According to Langmuir for a unimolecular layer, the following relation holds for adsorption

$$\frac{x}{m} = \frac{k_1 k_2 c}{1 + k_1 c}$$

where k_1 and k_2 are constant. The above equation can be written as

$$\frac{c}{x/m} = \frac{1}{k_1 k_2} + \frac{c}{k_2}$$

Thus a plot of $\frac{c}{x/m}$ against c gives a straight line with a slope of $\frac{1}{k_2}$ and intercept of $\frac{1}{k_1 k_2}$ and shows the validity of Langmuir's adsorption isotherm.

Scanned with CamScanner

Procedure :

1. Prepare 0.5 N acetic acid and 0.1 N Sodium hydroxide.
2. Take six clean and dry stoppered bottles. Prepare the following solution in each bottle. Prepare the following table.

Battle No.	0.5 N acetic acid (ml)	Distilled water (ml)	Amount of charcoal (gm)
1	50	0	1.0
2	40	10	1.0
3	30	20	1.0
4	25	25	1.0
5	20	30	1.0
6	10	40	1.0

3. Stopper each bottle and shake for at least one hour. And allow them to stand in a water bath for 30 minutes.
4. Filter each solution through a filter paper and collect the filtrate in numbered beakers. Reject the first 5 ml of each filtrate.
5. Pipette out 10 ml of each filtered solution in a conical flask and titrate it with 0.1 N NaOH solution, using phenolphthalein as indicator. Repeat titration with each solution until you get two concordant reading for each solution.
6. Note down the temperature of the water bath.
7. Titrate the stock solution of acetic acid (10 ml) also by means of 0.1 N NaOHsolution. 10 ml of stock acetic acid solution ≡ x ml of 0.1 N NaOH solution.

Bottle No.	Initial concentration (Co) of acid before adsorption	Equilibrium concentration (Ce) of acid after adsorption	Amount of acid (Co-Ce) adsorbed (ml of NaOH)
1	X (C_0)	C_e – from titration of B1	(C_0 –Ce) = x
2	4/5 X (C_0)	C_e – from titration of B2	(4/5C_0 –Ce) =
3	3/5 X (C_0)	C_e – from titration of B3	(3/5C_0 – Ce) =
4	1/2 X (C_0)	C_e – from titration of B4	(1/2 C_0 –Ce) =
5	2/5 X (C_0)	C_e – from titration of B5	(2/5 C_0 –Ce) =
6	1/5 X (C_0)	C_e – from titration of B6	(1/5 C_0 –Ce) =

Calculation : x = C_o – C_e and m = 1 gm in each case. Calculate log x/m and log C_e in each case.

Plot log x/m against log C_e. It is a straight line with a slope equal to $\frac{1}{n}$. This proves the validity of Freundlich adsorption isotherm.

To prove the validity of Langmuir adsorption isotherm the values of $\frac{Ce}{x/m}$ are plotted against C_e. It the adsorption isotherm is obeyed, the resultant curve will be a straight line.

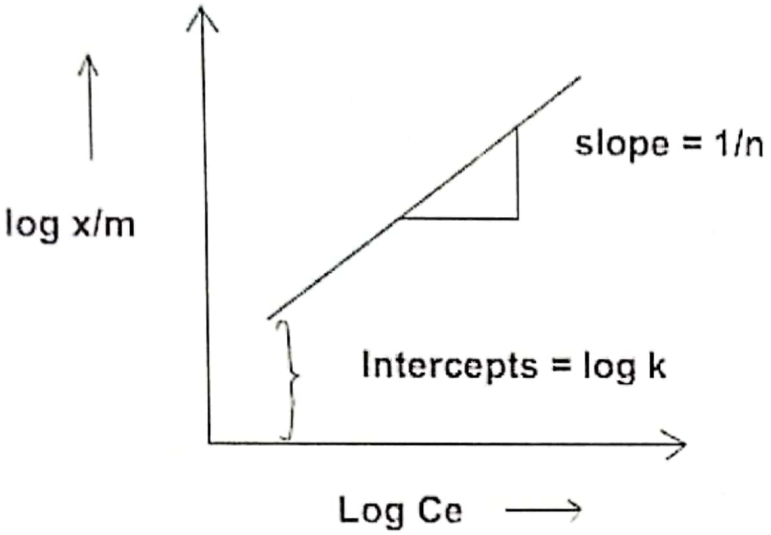
Real strength of acetic acid:

10 ml acetic acid x S_1 = 0.1M NaOH x Vol. NaOH

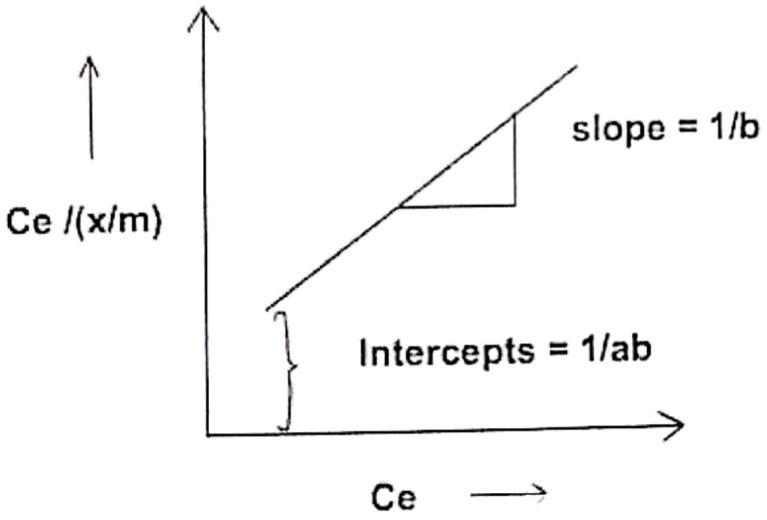
S_1 = C_0(X) initial con. ofacetic acid

Scanned with CamScanner

Freundlich Adsorption Isotherm -



Langmuir Adsorption Isotherm–



Standardize

Scanned with CamScanner

To study the liquid-liquid miscibility of three liquid system and to determine the position of one tie line

Apparatus and Chemicals required

Five stoppered bottles, burette, chloroform(dichloromethane), water

Theory: Liquid systems of three components consist of three liquid components, two liquids and one solid or one liquid and two solids. In this experiment, we investigate the behavior of a system of three liquids: acetic acid, chloroform, and water. Acetic acid is miscible with both chloroform and water, but chloroform and water are quite insoluble in each other.

According to the phase rule, a single phase ina three-component system may possess four degrees of freedom.

$$F = C - P + 2 = 3 - 1 + 2 = 4$$

F≡ degree of freedom ;C≡component ; P≡ phase

These are temperature, pressure and the compositions of two of the three components. Because of the difficulty in graphically so many variables, temperature and pressure are generally held constant, so the phaserule reduces to $F = C - P = 3 - P$.

This is the same special form of the phase rule that applies to two component systems of constant pressure only.

Graphical Representation of a Three-Component System

For a three–component system, the properties of an equilateral triangle provide a convenient means of representing the composition of a three–component system at constant temperature and pressure. Line AB represents the compositions of all systems that can be prepared from pure A and pure B: at point P the system is 60% B and 40% A.

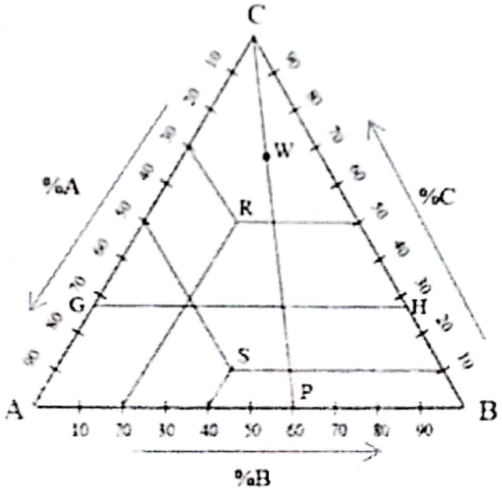
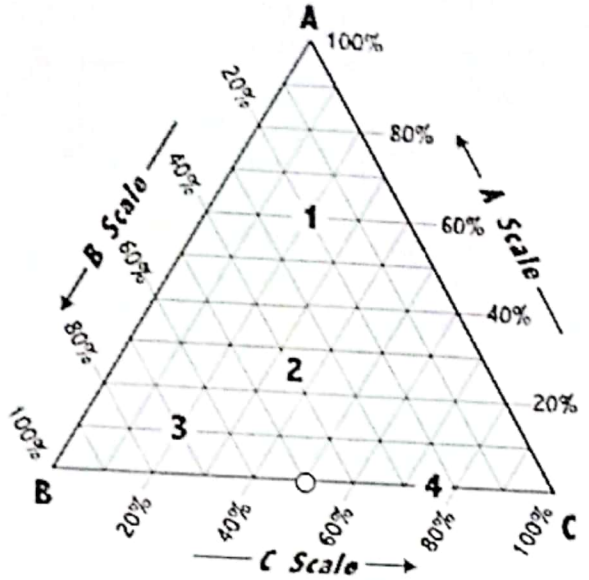
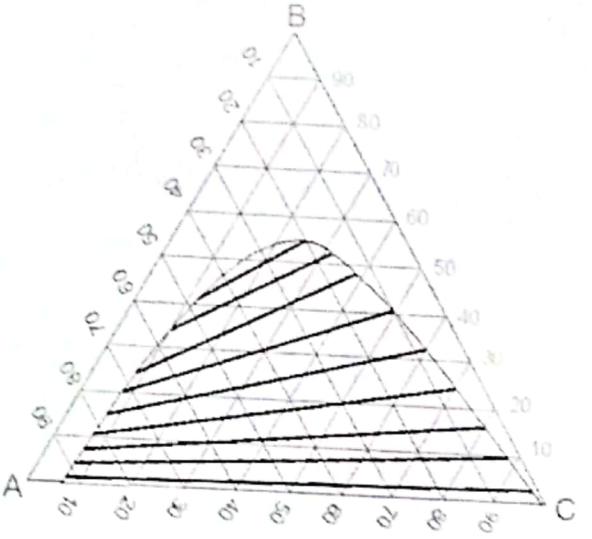


Fig. 1 Graphical representation of a three-component system at constant pressure and temperature

In Fig. 1, the edges of the triangle represent the compositions of all the two-component systems that can be prepared from A and B, B and C, and A and C. At point R the system is 50% C, 20% B, and 30% A. At point S the system is 10% C, 40% B, and 50% A.

Scanned with CamScanner





Procedure

1. Make 20gm of mixtures of $CHCl_3$ and CH_3COOH of varying composition in a set of nine stoppered bottles with 10, 20, 30, 40, 50, 60, 70,80, and 90% chloroform by weight. Since the densities of chloroform and acetic acid are 1.50 and 1.05 respectively, the mixtures may be prepared by running into the bottles from two burettes.
2. Now add water into these mixtures with constant shaking until turbidity appears.
3. From the amount of water added and the initial amounts of chloroform and acetic acid, the percentage composition of the mixture is calculated and the results are plotted in a triangular diagram.
4. Note the room temperature.

Chloroform		acetic acid		water
(g)	(ml)	(g)	(ml)	
2	1.33	18	17.14	2
4	2.67	16	15.24	
6	4.00	14	13.33	
8	5.33	12	11.43	
10	6.67	10	9.52	
12	8.00	8	7.62	
14	9.33	6	5.71	
16	10.67	4	3.81	
18	12.00	2	1.90	

For 10% chloroform, 90% acetic acid - bottle

% chloroform	1.33*100/(1.33+17.14+2)=	6.50
%acetic acid	17.14*100/(1.33+17.14+2)=	83.73
%water	2*100/(1.33+17.14+2)=	9.77

Scanned with CamScanner

Experiment 4

To determine the partial molal volume of solutions (Alcohol-Water system) by Westphal balance.

Theory: The partial molal property of a component of a solution may be defined as follows : Let G represent any extensive property of a binary solution ; at constant temperature and pressure, G then will be a function of the two independent variables n_1 and n_2 which represent the number of moles of component 1 and component 2 respectively. The partial molal quantity of component 1 is then defined by the relation –

$$\overline{G}_1 = \left(\frac{\delta G}{\delta n_1}\right) n_2,\ T, P \qquad \ldots\ldots\ldots (1)$$

Similarly for component 2,

$$\overline{G}_2 = \left(\frac{\delta G}{\delta n_2}\right) n_1,\ T, P \qquad \ldots\ldots\ldots (2)$$

Since any extensive property G of a mixture is a homogenous function of degree one of the two independent variables (n_1 and n_2), it follows from Euler's theorem (vide Defferential calculus) that –

$$G\ (n_1 \text{ and } n_2) = n_1 \overline{G}_1 + n_2 \overline{G}_2 \ \text{(T, P constant)} \qquad \ldots\ldots\ldots (3)$$

In the case of the volume of the solution by similar reasoning we have –

$$V\ (n_1 \text{ and } n_2) = n_1 \overline{V}_1 + n_2 \overline{V}_2 \ \text{(T, P constant)} \qquad \ldots\ldots (4)$$

Where $\overline{V}_1$ and $\overline{V}_2$ are the respective partial molal volumes.

For determining $\overline{V}_1$ and $\overline{V}_2$ the method of intercepts is convenient. Suppose we have mixtures where the total number of moles is one. For such systems, $n_1 + n_2 = 1$.

So, $dn_1 + dn_2 = 0$. Also, n_1 and n_2 are equal to the respective mole-fractions, N_1 and N_2, i.e. $n_1 = N_1$ and $n_2 = N_2$. Since $V = f(n_1, n_2)$, by differentiation,

$$dV = \left(\frac{\delta V}{\delta n_1}\right) dn_1 + \left(\frac{\delta V}{\delta n_2}\right) dn_2 = \overline{V}_1 dn_1 + \overline{V}_2 dn_2$$

or, $dV = -\overline{V}_1 dn_2 + \overline{V}_2 dn_2$

$$= dn_2 (\overline{V}_2 - \overline{V}_1)$$

$$\therefore \left(\frac{dV}{dn_2}\right) = \left(\frac{dV}{dN_2}\right) = (\overline{V}_2 - \overline{V}_1) \qquad \ldots\ldots\ldots (5)$$

This means that if the total volume of a system of two components is plotted against mole-fraction, the slope (tan a) at any point is equal to $(\overline{V}_2 - \overline{V}_1)$. If the total molality is also unity (i.e. $n_1 + n_2 = 1$) as shown in Fig. 1, it can be easily shown that the intercept, AC on the $n_1 = 1$ axis (i.e. $n_2 = 0$, cr pure A) is equal to $\overline{V}_1$ and the intercept BD on the $n_2 = 1$ axis (i.e. pure B) is equal to $\overline{V}_2$. This is deduced below :

Scanned with CamScanner

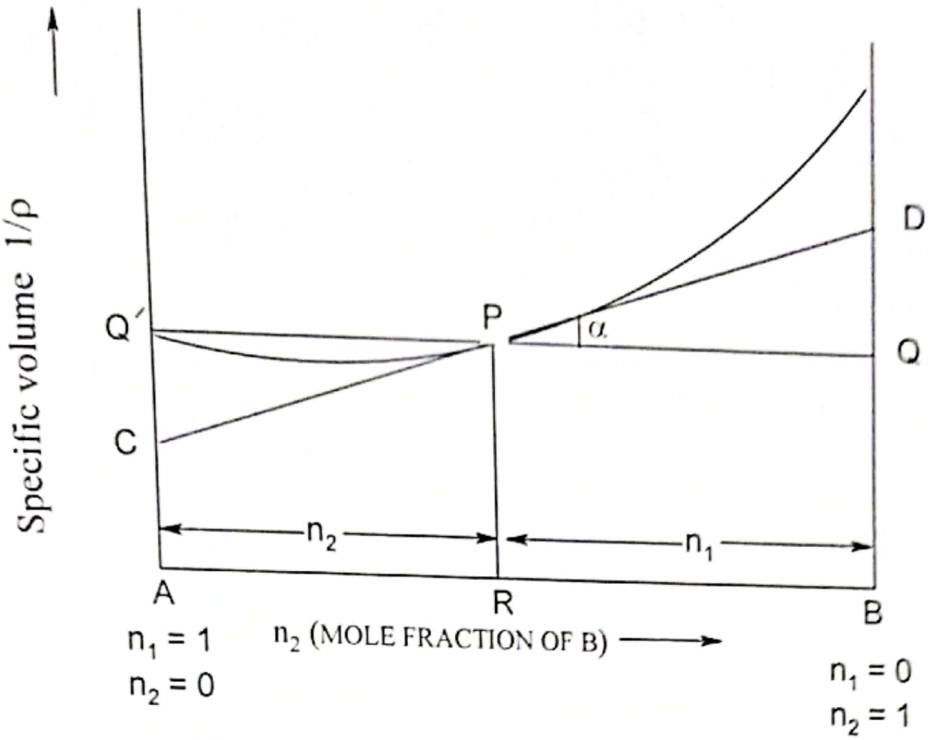


Fig. 1 : Specific Volume *versus* Mole-fraction.

If the volumes of mixtures of different compositions, but of a total molality of unity are plotted against mole fractions, then the intercepts BD and AC (Fig. 1) of the tangent to the curve at mole fraction n_2 are respectively equal to $\overline{V}_2$ and $\overline{V}_1$ of a solution of composition n_2, i.e., corresponding to R. This is evident from the following considerations: Let α be the angle between the tangent and the line V=0. The slope of the tangent is

$$\tan\alpha = \left(\frac{dV}{dn_2}\right) = \overline{V}_2 - \overline{V}_1$$

Now,

$$\begin{aligned} BD &= BQ + QD = PR + PQ \tan\alpha \\ &= V + n_1 \tan\alpha \text{ (here, } V \text{ is the total volume of composition, } n_2 : n_1) \\ &= \overline{V}_1 n_1 + \overline{V}_2 n_2 + n_1(\overline{V}_2 - \overline{V}_1) \\ &= \overline{V}_2 (n_2 + n_1) \\ &= \overline{V}_2 \end{aligned}$$

Similarly,

$$\begin{aligned} AC &= AQ' - CQ' = PR - PQ' \tan\alpha \\ &= V - n_1 \tan\alpha \\ &= \overline{V}_1 n_1 + \overline{V}_2 n_2 - n_2(\overline{V}_2 - \overline{V}_1) \\ &= \overline{V}_1 (n_1 + n_2) \\ &= \overline{V}_1 \end{aligned}$$

In practice, the reciprocal of density is plotted against weight percent; the intercept of the tangent to the curve at any concentration gives the partial specific volume at that concentration; When this is multiplied by the molecular weight of the component, one gets the partial molal volume. Alternatively, the molal volume could be plotted against the mole fraction and the method of intercept applied to get the partial molal volume directly.

Scanned with CamScanner

Materials

1. Methanol
2. Distilled water

Apparatus and Chemicals required

Westphal balance, alcohol

Procedure

1. Prepare a series aqueous alcohol solutionsby taking 15, 30, 45, 60, 75, 90% (wt/wt) of methanol.
2. Determine the density of distilled water ρ_w ,alcohol$\rho_{alc}$and each solutions by Westphal balance.
3. Calculate the specific volume ($1/\rho$) of each solution, including that of pure methanol and distilled water.
4. Plot the % by weight of ethanol (abscissa) against the specific volume. Draw a smooth curve through the points and draw tangents to curve at different concentration. The intercepts of these lines on the 0% and 100% ordinates gives the partial specific volumes of water and ethanol respectively at various concentrations.
5. Calculate the partial molal volume, i.e., partial specific volume x molecular weight.
6. Show the variation of (a) the partial molal volume of ethanol and (b) the partial molal volume of water with mole fraction.

Calculations and Results

Ethanol mole fraction	Density, ρ (gm./ml.)	Specific volume 1/ρ (ml./gm.)

Fig. 1 shows the nature of the plot. Calculate the partial molal volume of ethyl alcohol at, say, $n_2=X_2=0.5$ by the intercept method.

Wt. lost in water = W_{H2O}

Wt. lost in sol. =W_{sol}

Density of solution, $\rho = \frac{W_{sol.}}{W_{H2O}} \rho_{H2O}$

Scanned with CamScanner

Experiment 5

To determine the partial molal volume of solutions (Alcohol-Water system) by pycnometer

Theory: Same as Exp. 4.



Materials

1. Alcohol, Pycnometer
2. Distilled water

Apparatus and Chemicals required

pycnometer, alcohol

Procedure

1. Prepare series of aqueous alcohol solutions by taking 15, 30, 45, 60, 75, 90% (wt/wt) of alcohol.
2. Measure the weight of the clean and dry pycnometer.
3. Fill it with water (solution A) accordingly and measure its weight.
4. After the measurement, clean the pycnometerand dry it.
5. Take weight of pycnometeragain by filling it with the pure alcohol(solution B).
6. Now take the weights of pycnometerfilling it with each solution of different composition.
7. Determine the exact volume of pycnometer using the density of water at the measured temperature by the following equation,

   $V = \frac{m_{H2O}}{d_{H2O}}$, where d and m are the density and mass of water.
8. Calculate the densities of different composition solution using the equation,

   $$d_l = \frac{m_l}{m_{H2O}} d_{H2O}$$

   Where m_l and m_{H2O} are the weights of solutions and water and d_{H2O} is the density of water.
9. Calculate the average molar mass ($M_{average}$) and molar volume V_{molar} .for each mixture using the following equations

   $M_{average} = X_A M_A + (1-X_A)M_B$

   $V_{molar} = M_{average} / \rho$
10. Plot the mole fraction of alcohol (abscissa) against the molar volume (V_{molar}). Draw a smooth curve through the points and draw tangents to curve at different concentration which will give partial molar volume of water and alcohol at the intercepts of these lines on the 0% and 100% ordinates of B.

Table-

	X_B	X_A $=1-X_B$	$M_{average}$ (g/mol)	Mass of pycnometer + liquid (g)	Mass of liquid(g)	Density (g/cm^3)	Molar volume (V_m)
(A)	0.00	1.00			w_{H2O}		
(1)					w_{sol}	$d_1 = (w_{sol}/w_{H2O})\, d_2$	
(2)							
(B)	1.00	0.00			w_{alc}		

Scanned with CamScanner

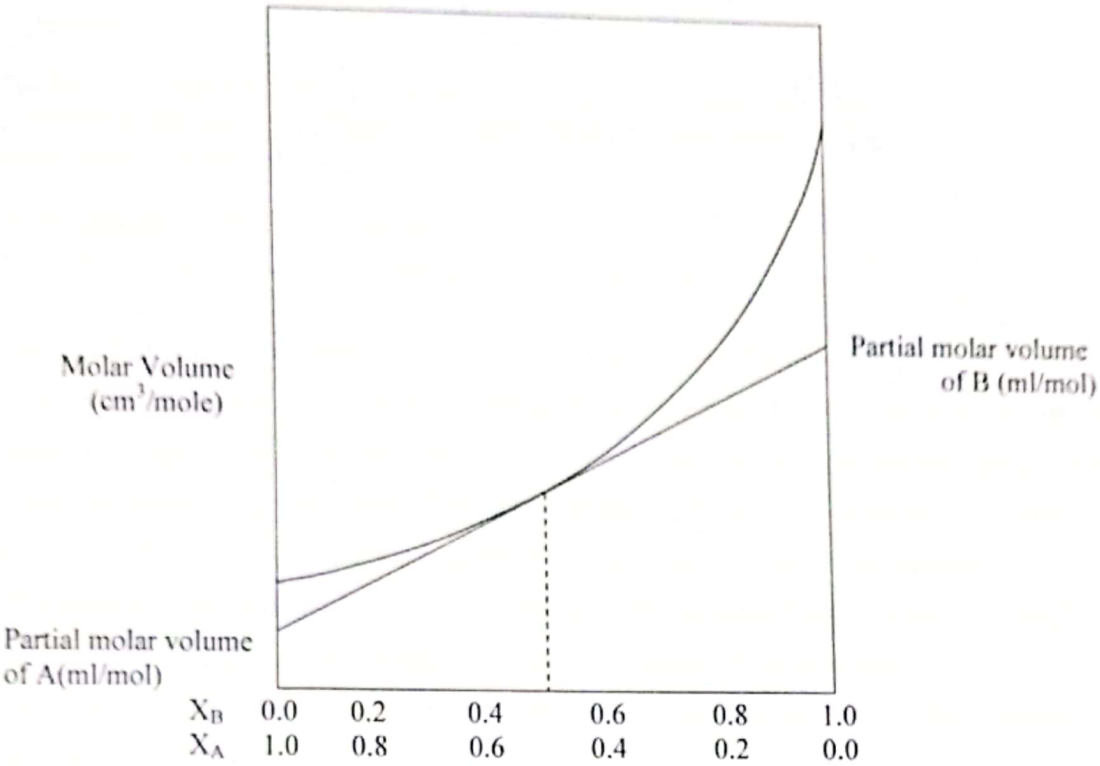


Fig. 1 shows the nature of the plot.

Calculate the partial molal volume of alcohol at, say, X_B=0.60 by the intercept method.

Alcohol (g)	no. of mol of alcohol	Water(g)	no. of mol of water	X_{alc}	X_{H2O}
15	15/32=0.47	85	85/18=4.72	0.09	0.91
30	30/32	70	70/18		
45	45/32	55	55/18		
60	60/32	40	40/18		
75	75/32	25	25/18		
90	90/32	10	10/18		

Mw. Ethanol = 46

Scanned with CamScanner

Experiment 6

To find (a) the specific rotation of sucrose (b) concentration of an unknown sucrose solution (C) Influence of solvent on the rotation of sucrose solution by polarimetric method

Apparatus and Chemicals required

Polarimeter, 100 ml volumetric flask, pipette, sucrose, 2N HCl

Theory: A solution of sucrose is optically active, that is, it has the ability to rotate the plane of polarization. It is dextro-rotatory, which means that it can rotate the plane of polarization of the plane polarized light in a clockwise manner. Fructose is laevo-rotatory, which means that it can rotate the plane of polarization of the plane polarized light in an anti-clockwise manner. Glucose is dextro rotatory, and the mixture of products has the resultant laevo-rotatory.

The angle of rotation depends on : (a) temperature ; (b) wavelength of light used; (c) length of the liquid column through which the light passes ; (d) nature of the substrate.

In order to indicate the rotatory power of the liquid or solution, we have to define a quantity known as SPECIFIC ROTATION.

The specific rotation of a substance (solute in solution) is given by :

$$[\alpha]_\lambda = 100\,\theta / c\,l$$

(a) Procedure for specific rotation

1. Prepare 100 ml each of 5%, 10%, 15%, 20% sucrose solution.
2. Clean the polarimeter tube and fill it in with distilled water. Place the circular glass piece (handle carefully) on the mouth of the tube and fit the screw with rubber washers. Do not screw too tightly. Take care so that no air bubble is trapped inside the tube.
3. Switch the sodium lamp on and place it in front of the polarimeter.
4. Focus the eyepiece of the telescope on the line dividing the field of view.
5. Determine the vernier constant of the vernier scale attached to the circular scale.
6. Place the polarimeter tube with water inside in between the polariser and the analyser, and rotate the analyser until the best match is obtained between the dark and bright parts of the field of view. Take reading on the circular scale. This is zero reading.
7. Use sucrose solution instead of water and determine the rotation for each of the solutions by repeating stops 2 and 6. Take care to rinse the polarimeter tube with appropriate solutions.
8. Subtract zero reading from the observed rotation for each solution.

   Calculate specific rotation using the formula

$$[\alpha]'_\lambda = \frac{\theta \times 100}{l \times C} = \text{specific rotation (deg cm}^2\text{g}^{-1}\text{)}$$

   where θ = rotation by solution at conc. Cg/100 ml, and l = length of the polarimeter tube in decimetre and $[\alpha]'_\lambda$ is the specific rotation of the substance at a given temperature and wavelength of light.

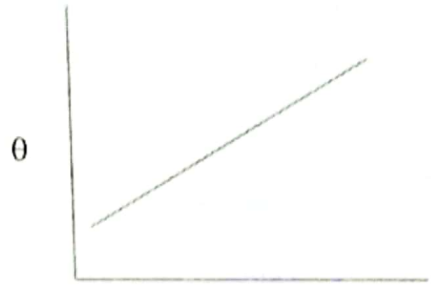
Scanned with CamScanner

(b) Procedure for unknown concentration

1. Plot rotation of 5%, 10%, 15%,20% sucrose solution against percept concentration.
2. Measure the angle of rotation of the unknown sucrose solution. The concentration corresponding to the angle of rotation read from the above plot gives the unknown concentration.

(C) Procedure for the determination of Influence of solvent

1. Take 50 ml of 10% cane sugar solution and add 50 ml acetone to it. Measure the angle of rotation and calculate the specific rotation. Compare the value for 5% solution of cane sugar.



Scanned with CamScanner

Experiment 7

To determine (a) the absorption curve of coloured substances in solution, (b) the validity of Beer-Lambert law, and (c) the strength of an unknown solution

Theory: Absorption spectroscopy can be used to quantify the absorbing species present in the sample. The greater the quantity of absorbing species present, the greater will be the extent to which the incident light will be absorbed. When a beam of monochromatic light falls on a substance, apart of it is absorbed and the rest is transmitted. The intensity of the transmitted light is decreased. According to Lambert's law, decrease in intensity (-dI) is proportional to the thickness of the medium (dl) and intensity of incident light (I). Beer extended this law towards solutions (and gases). In such cases decrease in intensity is also proportional to molar concentration (C).

So,

$$-dI = K.\ I.\ dl.\ C \text{ (K is a constant)}$$

Beer - Lambert's law states that for a solution, the absorbance of the sample (A) is given by:

$$A = \log (Io/I) = \varepsilon C\ l = O.\ D.$$

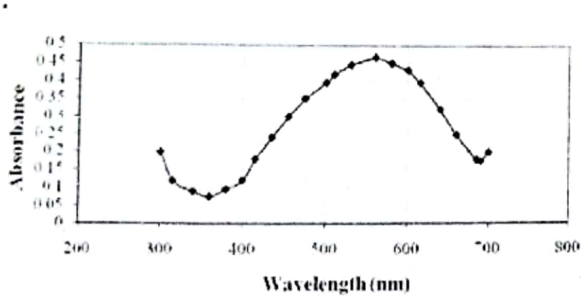
Where,

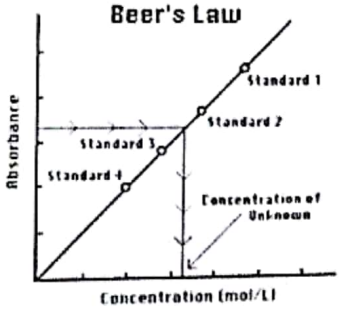
Io is the intensity of the incident light, I is the intensity of the transmitted light,
ε is the molar absorptivity (characteristic of the absorbing species),
C is the molar concentration of the absorbing speciesin the sample,
l is the path length, log (Io / I) = Optical density or Absorbance

Since ε depends on wave length of the light and nature of the substance so for a given substance at a given frequency and in a given photo chemical cell (l = constant), O.D. = constant * C.

Hence, a plot of O.D. Vs C will be a straight line through origin. This is Beer's law.

Absorbance has no units. If 'l' is expressed in cm, and 'C' in mol dm^{-3}, then 'ε' has the units of $mol^{-1}dm^3\ cm^{-1}$. A linear plot of absorbance versus concentration through origin would verify the Beer-Lambert's Law as shown below:





Scanned with CamScanner

Apparatus and Chemicals required

Spectronic 20, five 100ml volumetric flask, KMnO

Procedure

1. Prepare 100 ml of M/1000 $KMnO_4$ solution.
2. Take 5, 10, 15, 20 and 25 ml of the solution into different 100ml volumetric flask and dilute each up to the mark with distilled water.
3. Determine the wavelength for maximum absorbance, λ_{max} by measuring the absorbance of any of the above solution at different wavelength and plotting a absorbance vs. wavelength graph. Use distilled water as blank.
4. Determine the absorbance for each of these five solutions at the λ_{max}.
5. Plot absorbance vs. conc. and examine the validity of Beer-Lambert's law.
6. The concentration of any unknown solution can be found out by measuring the absorbance at the λ_{max} and using the above plot.
7. Repeat the above experiment using 1M H_2SO_4 instead of water as solvent.
8. Repeat the experiment using M/100 $K_2Cr_2O_7$ solution in water as well as in 1M H_2SO_4.

Result : It is observed that $KMnO_4$ solution in water and in acid obeys Beer-Lambert's law.

units of molar absorptivity are L mol^{-1} cm^{-1}

Scanned with CamScanner

Experiment 8

To determine the specific reaction rate of base catalysed hydrolysis of an ester by conductometric method

Theory: In the presence alkali ethyl acetate undergoes hydrolysis in accordance with

$$CH_3COOC_2H_5 + OH^- = CH_3COO^- + C_2H_5OH$$

The rate of saponification is directly proportional to both concentrations of ester and alkali, and the reaction therefore, is a second order one, i.e.

$$k = \frac{1}{t}\frac{x}{a(a-x)} \qquad \text{or,} \qquad C_t = \frac{1}{ak}\frac{C_0 - C_t}{t} + C_\infty$$

where a and b are the initial concentrations of ester and alkali, respectively (a = b in this case) and k is the reaction rate constant.

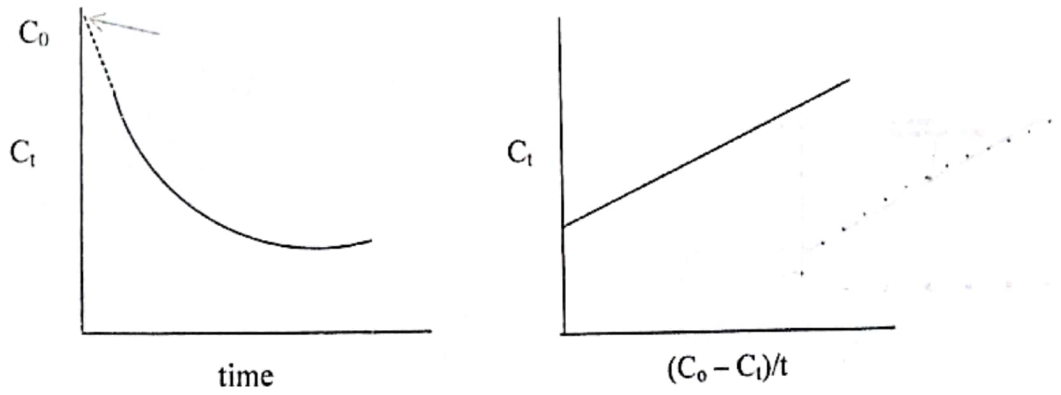
Apparatus and Chemicals required

Conductometer and cell, thermostat bath, 250 ml conical flask, pipette 0.1M NaOH, 0.1M Ethyl acetate or 0.1M Methyl acetate

Procedure

1. Prepare 100 ml of exactly 0.1M ester solution (at the time of experiment).
2. Prepare exactly 0.1M NaOH solution.
3. Place them in a constant thermostat both at 30°C.
4. Wash the conductivity cell with distilled water.
5. Transfer 25 ml of NaOH solution into a beaker or large test tube and carefully and 25 ml of ester to it.
6. Immediately start the stop-clock.
7. Measure the conductance at known intervals (C_t).
8. Plot C_t vs. time and find out the conductance value at t = 0 by extrapolating, this is C_o.
9. Plot C_t conductance at time t against $(C_o - C_t)/t$ and determine its slope to calculate the rate constants.

Slope = $\frac{1}{ak}$ where a is the initial concentration of ester and k is the specific reaction rate.



Scanned with CamScanner

Experiment 9

To find out the equivalent conductance at infinite dilution and the dissociation constant of a weak electrolyte

Apparatus and Chemicals required

Conductometer and cell, beaker, stirrer, N/10 acetic acid

Theory: weak electrolytes dissociate as follows

$$CH_3COOH \leftrightarrows CH_3COO^- + H^+$$

Degree of dissociation $\alpha = \frac{\Lambda_c}{\Lambda_0}$

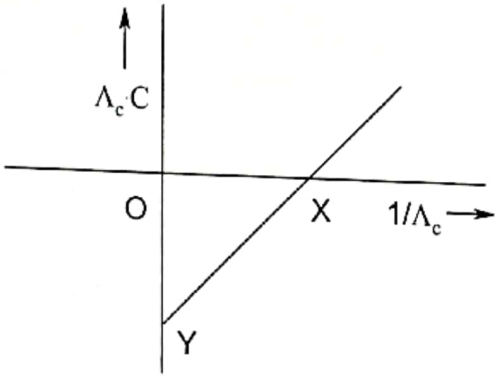
According to Ostwald's dilution law

$$K_c = \frac{\alpha^2 C}{1-\alpha}$$

$$= \frac{\Lambda_c^2 . C}{\Lambda_0(\Lambda_0 - \Lambda_c)}$$

$$\frac{\Lambda_0}{\Lambda_c^2} = K_c \frac{(1 - \Lambda_c / \Lambda_0)}{\Lambda_c . C}$$

$$\Lambda_c.C = K_c\left(\frac{\Lambda_0^2}{\Lambda_c} - \Lambda_0\right) = K_c \frac{\Lambda_0^2}{\Lambda_c} - K_c\Lambda_0$$



Plot of $\Lambda_c.C$ against $\frac{1}{\Lambda_c}$ will thus give a straight line.

At $\Lambda_c.C = 0$, $OX = \frac{1}{\Lambda_0}$

that is $\Lambda_0 = \frac{1}{OX}$

At $\frac{1}{\Lambda_c} = 0$, $OY = -K_c.\Lambda_0$

Therefore $(OX).(OY) = -K_c$

Procedure

1. Prepare 250 ml of approx. N/10 Acetic acid solution.
2. Standardise the acid solution with standard sodium hydroxide solution.
3. From the stock solution, prepare acid solution of the following strengths N/20, N/30, N/200, N/400, N/1000, N/2000, N/10000.
4. Determine the conductance for each of the solution as in the earlier experiment.
5. Measure the conductance of distilled water used in the experiment and subtract it from the observed conductance of each solution.

Scanned with CamScanner

6. Calculate the equivalent conductance (Λ_c) of each solution, Plot Λ_c vs. $\sqrt{conc.}$ and comment on the nature and cause of this plot.
7. Assume $\Lambda_o = 390.8$ mho equiv^{-1} cm^2 for acetic acid and calculate degree of dissociation for each solution from $\alpha = \Lambda_c/\Lambda_o$
8. Substitute for α in the Ostwald's dilution law, $K_a = \alpha^2C/(1-\alpha)$ and plot Λ_c vs. $1/\Lambda_c$ and calculate for the dissociation constant.
9. Calculate the values of dissociation constant K_a for each solution and comment on the magnitude of K_a.

Sol. Con.	Conductance(κ)	$\Lambda_c = 1000\kappa/C$	$\Lambda_c C$	$1/\Lambda_c$	$\alpha = \Lambda_c/\Lambda_0$	$K = \alpha^2 / (1-\alpha^2)$	$\sqrt{C}$
N/20							
N/30							
N/200							
N/400							
N/1000							
N/2000							
N/10000							

(1a)	electrical conductivity (specific conductance)	EC	in S/m (or µS/cm)
(1b)	molar conductivity	$\Lambda_m = EC / c$	in S m^2 mol^{-1}
(1c)	equivalent conductivity	$\Lambda_{eq} = \Lambda_m / \|z\|$	in S cm^2 eq^{-1}

Scanned with CamScanner

Experiment 10

Conductometric titration of (a) strong and (b) weak acids and to draw their neutralisation curves

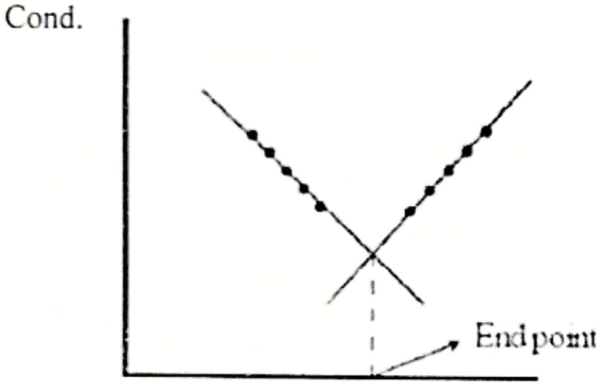
Theory: Conductometric titration is the volumetric analysis, based upon the measurement of theconductance during the titration. The number of ions, charge on the free ion and mobility of theions effect the conductance of a solution. When one electrolyte is added to another electrolyte, thechange in number of free ions causes a change in the conductance.

For example NaOHVsHCl: Before adding NaOH solution to HCl, HCl has high conductivitydue to highly mobile H^+ions. When NaOH is added, the conductivity of HCl decreases due toneutralization of highly mobile H^+ions by OH^-ions of NaOH.

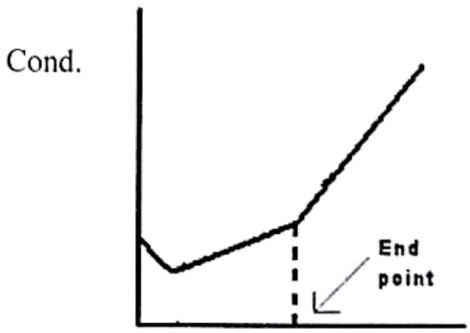
$$H^+Cl^- + Na^+OH^- \rightarrow Na^+ + Cl^- + H_2O$$

The conductivity of solution decreases until an equivalent point is reached. Further addition of NaOH increases the conductivity of solution due to increase in the highly mobile OH-ions.

strong acid



weak aid



Weak Acid with a Strong Base, e.g. acetic acid with NaOH: Initially the conductance is low due to the feeble ionization of acetic acid. On the addition of base, there is decrease in conductance not only dueto the replacement of H^+byNa^+but also suppresses the dissociation of acetic acid due to common ion acetate. But very soon, the conductance increases on adding NaOH as NaOH neutralizes the un-dissociated CH_3COOH to CH_3COONa which is the strong electrolyte. This increase in conductance continuesraise up to the equivalence point. The graph near the equivalence point is curved due the hydrolysis of salt CH_3COONa. Beyond the

Scanned with CamScanner

equivalence point, conductance increases more rapidly with the addition of NaOH due to the highly conducting OH^- ions.

Apparatus and Chemicals required

Conductometer and cell, beaker, volumetric flask, HCl, acetic acid, NaOH

Procedure for strong acid

1. Prepare 0.2N NaOH solution. Fill in the burette with 0.2N NaOH and standardise with standard 0.1N succinic acid.
2. Pipette out 10.0 ml of 0.1N HCl solution (approx.) into a clean beaker and dilute to 100 ml with distilled water.
3. Add 1 ml portion of the 0.2N NaOH soln. from the burette to the acid soln. After each addition (i) note the burette reading (ii) stir the soln. gently and then let it stand for a minute (iii) measure the conductance of the mixture. Continue the titration, till the end point indicated by a fall and then rise in conductance is reached. This is your rough titration. Plot conductance vs. volume of alkali added graph and locate the end point.
4. Perform a fair titration in the same manner, now adding 0.1 ml portions of alkali near the end point. Take care to wash and rinse the conductance cell and beaker with distilled water before starting the fair titration.
5. Plot conductance vs. the volume of NaOH and obtain the neutralisation curve more accurately.
6. Find out the volume of NaOH required to neutralise the HCl solution and calculate the strength of HCl

Procedure for weak acid

1. Perform step one of the above procedure.
2. Pipette out 10.0 ml of 0.1N acetic acid soln. (approx.) into a clean beaker and dilute to 100 ml with distilled water.
3. Stir the solution allow to stand and measure the conductance of solution.
4. Add 0.1 ml portion of the 0.2N NaOH from the burette. After each addition (i) note the burette reading (ii) Stir the solution gently and then let it stand for a minute (iii) measure the conductance of the solution. Continue this process until about 0.5 ml NaOH solution has been added. Explain the cause of this slow addition of NaOH at the beginning of titration.
5. Next add 1 ml portion of the NaOH solution to the acid and after each addition continue noting the burette reading, stirring the solution and taking the conductance reading. Continue the titration, till the end point, indicated by a slow and then more rapid rise in conductance vs. volume of alkali added graph and locate the end point.
6. Perform a fair titration in the same manner now adding 0.1 ml portions of alkali near the end point.
7. Plot conductance vs. volume of NaOH and obtain the neutralisation curve more accurately. Explain the difference in the neutralisation curves between the strong and weak acid.

Scanned with CamScanner

Experiment 11

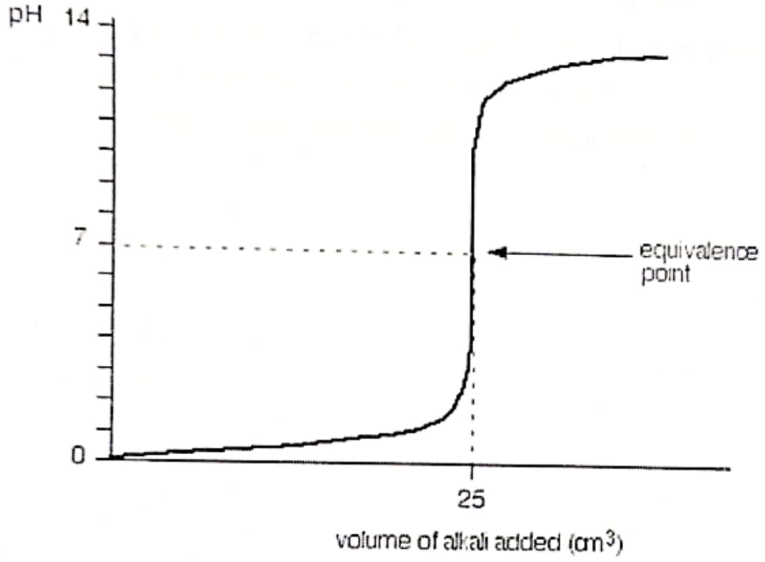
Potentiometric titration of (a) strong and (b) weak acids by a strong base and to draw their neutralisation curves

Theory: A reaction, in which hydrogen ions are produced or consumed, can be followed by ob serving the pH of the reaction. Negative logarithm of the concentration of the H+ cations gives the pH of the solution.

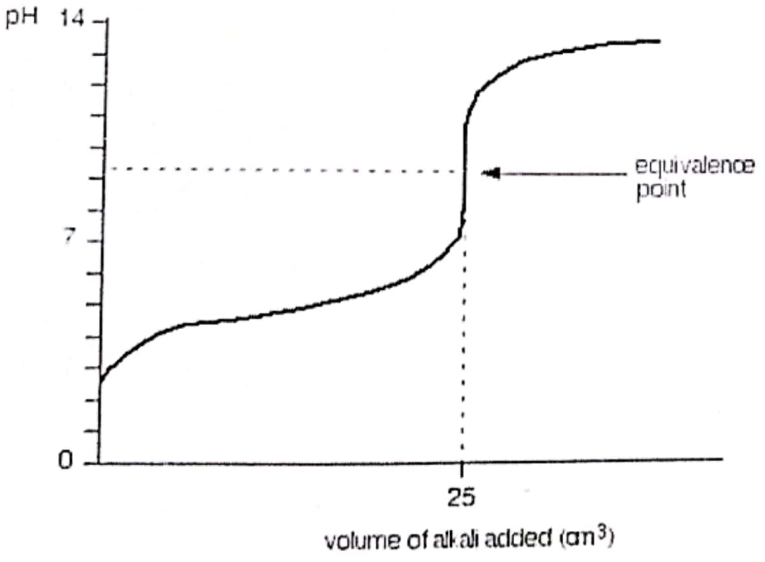
$pH = - \log[H+]$

Potentiometric titration can serve to determine the equivalence point or end point of titration. As the titration is carried out, as long as the acid is in excess, the pH will vary slightly with addition of base. However, near the equivalence point, concentration of the H+ cations fall rapidly.

strong acid



weak aid



Scanned with CamScanner

Apparatus and Chemicals required

pH meter, volumetric flask, beakers, strong acid, weak acid, 0.1N KCl solution, standard 0.1N NaOH solution, quinhydrone.

Procedure for potentiometric titration

1. Fill in the burette with standard 0.1N NaOH solution.
2. Pipette out 10 ml of 0.1N (approx.) HCl solution into a clean beaker and dilute to 100 ml with distilled water. Put a pH electrode in to the beaker.
3. Add 1ml portion of the 0.1N NaOH solution from a burette to the acid solution. After each addition (i) note the burette reading (ii) stir the solution gently and (iii) measure the pH of the solution. Continue the titration till the end point, indicated by a sharp change in the value of pH of the solution. Take few more readings. This is your rough titration. Plot pH vs. volume of alkali added graph and locate the end point.
4. From the pH vs. volume curve, obtain values of $\Delta pH/\Delta V$ at various mean values of V. Tabulate the derived data properly for this purpose.
5. Plot $\Delta pH/\Delta V$ vs. mean volume, and locate the end point.
6. Note the pH of the solution at the end point.
7. Repeat the experiment with the acetic acid solution.

Scanned with CamScanner

Experiment 12

To determine the standard electrode potential of zinc and copper

Apparatus and Chemicals required

Potentiometer, standard cell, galvanometer, saturated calomel electrode, zinc electrode, copper electrode, 0.1M solution $ZnSO_4$ and $CuSO_4$

Theory: The saturated calomel electrode (SCE) is connected with copper or zinc electrode and the e.m.f of the cell is determined by means of a potentiometer. Thus

$$E_{cell} = E_{cathode(red)} - E_{anode(red)}$$

Two cells can be represented as:

$$Zn, Zu^{2+}(aq.) \parallel \text{Calomel (saturated)}$$
$$\text{Calomel (saturated)} \parallel Cu^{2+}(aq.), Cu$$

The standard electrode potential (E^0) is given by

$$E = E^0 - \frac{2.303RT}{nF} logQ$$

Procedure

1. Prepare 100 ml of 0.1M $ZnSO_4$ and 0.1M $CuSO_4$ solutions.
2. Prepare 0.01M, 0.001M, 0.0001M $ZnSO_4$ and $CuSO_4$ solutions by diluting 0.1M, 0.01M 0.001M solutions, respectively.
3. Set up the cells as
   (i) Calomel (saturated) $\parallel Cu^{2+}$(aq.), Cu, (ii) Zn, Zu^{2+}(aq.) $\parallel$ Calomel (saturated)
4. Set up a potentiometer and standardised it against a standard cell with no deflection at the attached galvanometer. Now connect the experimental cell (calomel $\parallel Cu^{2+}(0.1)$, Cu) to the potentiometer and measure its e.m.f. Measure the e.m.f. of cell using solutions of different concentrations of $CuSO_4$ each time. Take at least three readings for each solution.
5. Repeat the step 4 for Zn electrode for $ZnSO_4$ solution.
6. Calculate E^o of each electrode by graphical method.
   Using the values of mean ionic activity coefficient given below, E^o can also be calculated using the following equation (1) and (2).

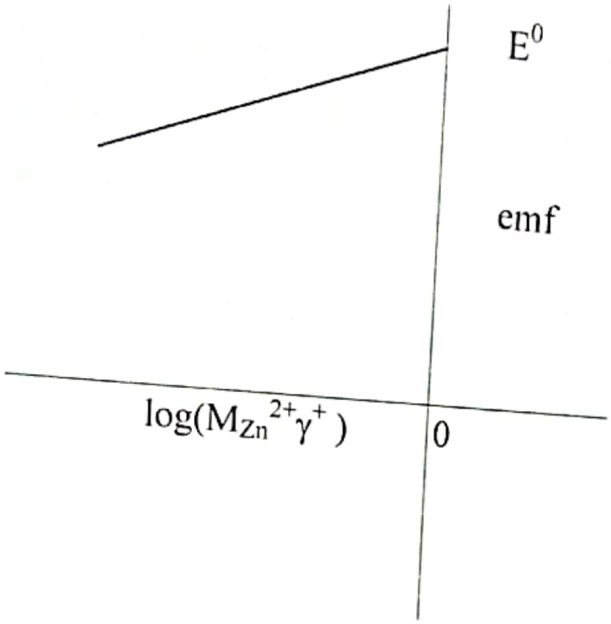
$$E_{cell} = E_{Zn/Zn^{2+}} + E_{Calomel(sat)} = E^o_{Zn/Zn^{2+}} - \frac{RT}{2F} \ln M_{Zn^{2+}} \cdot \gamma_+ + E_{Calomel(sat)} \quad (1)$$

$$E_{cell} = E_{Calomel(sat)} + E_{Cu^{2+}/Cu} = E_{Calomel(sat)} + E^o_{Cu^{2+}/Cu} + \frac{RT}{2F} \ln M_{Cu^{2+}} \cdot \gamma_+ \quad (2)$$

The values of γ_+ at 25°C are:

	0.1M	0.01M	0.001M	0.0001M
$CuSO_4$	0.158	0.404	0.432	0.437
$ZnSO_4$	0.161	0.400	0.431	0.432

Scanned with CamScanner



Standardize

$E°_{Cu2+/Cu}$=0.339V (reduction)
$E°_{Zn2+/Zn}$= -0.76 V (reduction)

Scanned with CamScanner

$$E_{cell} = E_{Zn/Zn^{2+}} + E_{Calomel(sat)}$$

$$E_{cell} = E^{o}_{Zn/Zn^{2+}} - \frac{RT}{2F} \ln M_{Zn^{2+}} \cdot \gamma_{+} + E_{Calomel(sat)}$$

$$E_{cell} = E_{Calomel(sat)} + E_{Cu^{2+}/Cu}$$

$$= E_{Calomel(sat)} + E^{o}_{Cu^{2+}/Cu} + \frac{RT}{2F} \ln M_{Cu^{2+}} \cdot \gamma_{+}$$

Scanned with CamScanner